Chapter 1. Introduction to Object Oriented Programming

What is C++?

C++ is a cross-platform language that can be used to create high-performance applications.

C++ was developed by Bjarne Stroustrup, as an extension to the C language.

C++ gives programmers a high level of control over system resources and memory.

C++ is a popular programming language.

C++ is used to create computer programs, and is one of the most used language in game development.

C++ was developed as an extension of C, and both languages have almost the same syntax.

Why Use C++

C++ is one of the world's most popular programming languages.

C++ can be found in today's operating systems, Graphical User Interfaces, and embedded systems.

C++ is an object-oriented programming language which gives a clear structure to programs and allows code to be reused, lowering development costs.

C++ is portable and can be used to develop applications that can be adapted to multiple platforms.

Example

```
#include <iostream>
using namespace std;

int main() {
  cout << "Hello World!";
  return 0;
}
```

2. Difference between C and CPP

C was developed by Dennis Ritchie between the year 1969 and 1973 at AT&T Bell Labs.	C++ was developed by Bjarne Stroustrup in 1979.
C is (mostly) a subset of C++	C++ is (mostly) a superset of C.
C supports procedural programming.	C++ supports both procedural and object oriented programming
C does not support information hiding.	Data is hidden by the Encapsulation to ensure that data structures and operators are used as intended.
C is a function-driven language.	C++ is an object-driven language
Namespace features are not present inside the C.	Namespace is used by C++, which avoid name collisions.
Standard IO header is stdio.h.	Standard IO header is iostream.h.
C does not support inheritance.	C++ supports inheritance.
Direct support for exception handling is not supported by C.	Exception handling is supported by C++.
scanf() and printf() functions are used for input/output in C.	cin and cout are used for input/output in C++.
C structures don't have access modifiers.	C ++ structures have access modifiers.
There are 32 keywords in the C	There are 97 keywords in the C++
File extension is ".c"	File extension is ".cpp"

Features of OOPL /OOPL Concepts/Characteristics of OOP Language

1) Simple

C++ is a simple language because it provides a structured approach (to break the problem into parts), a rich set of library functions, data types, etc.

2) Portable

C++ is a portable language and programs made in it can be run on different machines.

3) Mid-level / Intermediate programming language

C++ includes both low-level programming and high-level language so it is known as a mid-level and intermediate programming language. It is used to develop system applications such as kernel, driver, etc.

4) Structured programming language

C++ is a structured programming language. In this we can divide the program into several parts using functions.

5) Rich Library

C++ provides a lot of inbuilt functions that make the development fast. Following are the libraries used in C++ programming are:

- <iostream>
- <cmath>
- <cstdlib>
- <fstream>

6) Memory Management

C++ provides very efficient management techniques. The various memory management operators help save the memory and improve the program's efficiency. These operators allocate and deallocate memory at run time. Some common memory management operators available C++ are new, delete etc.

7) Object-Oriented

In C++, object-oriented concepts like data hiding, encapsulation, and data abstraction can easily be implemented using keyword class, private, public, and protected access specifiers. Object-oriented makes development and maintenance easier.

8) Reusability

With the use of inheritance of functions programs written in C++ can be reused in any other program of C++. You can save program parts into library files and invoke them in your next programming projects simply by including the library files. New programs can be developed in lesser time as the existing code can be reused. It is also possible to define several functions with same name that perform different task.

9) Extensible

C++ programs can easily be extended as it is very easy to add new features into the existing program.

- C++ Classes/Objects

C++ is an object-oriented programming language.

Everything in C++ is associated with classes and objects, along with its attributes and methods. For example: in real life, a car is an object. The car has attributes, such as weight and color, and methods, such as drive and brake.

Attributes and methods are basically variables and functions that belongs to the class. These are often referred to as "class members".

A class is a user-defined data type that we can use in our program, and it works as an object constructor, or a "blueprint" for creating objects.

Create a Class

To create a class, use the class keyword:

Example

Create a class called "MyClass":

```
class MyClass {       // The class
  public:             // Access specifier
    int myNum;        // Attribute (int variable)
    string myString;  // Attribute (string variable)
};
```

- The class keyword is used to create a class called MyClass.
- The public keyword is an access specifier, which specifies that members (attributes and methods) of the class are accessible from outside the class. You will learn more about access specifiers later.
- Inside the class, there is an integer variable myNum and a string variable myString. When variables are declared within a class, they are called attributes.
- At last, end the class definition with a semicolon ;.

Create an Object

In C++, an object is created from a class. We have already created the class named MyClass, so now we can use this to create objects.

To create an object of MyClass, specify the class name, followed by the object name.

To access the class attributes (myNum and myString), use the dot syntax (.) on the object:

C++ OOPs Concepts

The major purpose of C++ programming is to introduce the concept of object orientation to the C programming language.

Object Oriented Programming is a paradigm that provides many concepts such as inheritance, data binding, polymorphism etc.

OOPs (Object Oriented Programming System)

Object means a real word entity such as pen, chair, table etc. Object-Oriented Programming is a methodology or paradigm to design a program using classes and objects. It simplifies the software development and maintenance by providing some concepts:

- Object
- Class
- Inheritance
- Polymorphism
- Abstraction
- Encapsulation

Object

Any entity that has state and behavior is known as an object. For example: chair, pen, table, keyboard, bike etc. It can be physical and logical.

Class

Collection of objects is called class. It is a logical entity.

A Class in C++ is the foundational element that leads to Object-Oriented programming. A class instance must be created in order to access and use the user-defined data type's data members and member functions. An object's class acts as its blueprint.

Take the class of cars as an example. Even if different names and brands may be used for different cars, all of them will have some characteristics in common, such as four wheels, a speed limit, a range of miles, etc. In this case, the class of car is represented by the wheels, the speed limitations, and the mileage.

Inheritance

When one object acquires all the properties and behaviours of parent object i.e. known as inheritance. It provides code reusability. It is used to achieve runtime polymorphism.

1. Sub class - Subclass or Derived Class refers to a class that receives properties from another class.
2. Super class - The term "Base Class" or "Super Class" refers to the class from which a subclass inherits its properties.
3. Reusability - As a result, when we wish to create a new class, but an existing class already contains some of the code we need, we can generate our new class from the old class thanks to inheritance. This allows us to utilize the fields and methods of the pre-existing class.

Polymorphism

When one task is performed by different ways i.e. known as polymorphism. For example: to convince the customer differently, to draw something e.g. shape or rectangle etc.

Abstraction

Hiding internal details and showing functionality is known as abstraction. Data abstraction is the process of exposing to the outside world only the information that is absolutely necessary while concealing implementation or background information.For example: phone call, we don't know the internal processing.

Encapsulation

Binding (or wrapping) code and data together into a single unit is known as encapsulation. For example: capsule, it is wrapped with different medicines.

Encapsulation is typically understood as the grouping of related pieces of information and data into a single entity. Encapsulation is the process of tying together data and the functions that work with it in object-oriented programming.

Dynamic Binding - In dynamic binding, a decision is made at runtime regarding the code that will be run in response to a function call. For this, C++ supports virtual functions.

Why do we need oops in C++?

There were various drawbacks to the early methods of programming, as well as poor performance. The approach couldn't effectively address real-world issues since, similar to procedural-oriented programming, you couldn't reuse the code within the program again, there was a difficulty with global data access, and so on.

With the use of classes and objects, object-oriented programming makes code maintenance simple. Because inheritance allows for code reuse, the program is simpler because you don't have to write the same code repeatedly. Data hiding is also provided by ideas like encapsulation and abstraction.

Applications of C++

1. Operating Systems

C++ was used in developing most of the operating systems in the world, from Microsoft Windows to macOS to Linux.

C++ is a strongly typed fast programming language that gives it an upper hand over other languages for developing various OS. Whereas C is a low-level language that developers use for developing drivers in the respective OS.

2. GUI Based Applications

C++ was also used in developing most of the GUI (Graphical User Interface) applications that we use regularly.

For example, C++ is the backbone for scripting GUI applications from Adobe i.e. Photoshop, Illustrator, Adobe Premier.

3. Browsers

Most of the browser requires a faster rendering process as the users do not like to wait for the page to load. Since C++ has faster execution, it helps in developing rendering engines in browsers.

C++ supported all the major browsers like Google Chrome, Mozilla (Firefox and Thunderbird), Safari, Opera, and many more.

4. Graphics

C++ has high performance and speed which comes in handy for making applications that require 3D animation, modeling, image processing, and real-time simulations. It is beneficial to use C++

as a back-end programming language for all the graphics applications that require faster loading and minimum rendering, just like in browsers.

These features helped develop mobile apps which use various different kinds of sensors, visual effects (VFX toolkit), and real-time image processing.

5. Database Software

Along with browsers and GUI applications, C++ is also used in developing the most important DBMS engines like MySQL, Postgres, Redis, and Oracle. These softwares are the backbone of many top-tier IT companies including Google, Wikipedia, and Yahoo.

Features like file handling, speed along OOP concepts like classes, objects, functions, and reliability makes C++ a preferable scripting language for developing this database software.

6. Libraries

C++ has got a great set of in-built standard libraries which help in developing high-level machine learning and mathematical computation libraries.

Most high-level machine language libraries use C++ as a backend programming language. Such libraries require huge calculations and operations in order to train Machine Learning models, for which C++ comes to help the developers.

7. Cloud computing

Nowadays cloud storage systems are extensively used in the current technology scenario. Cloud system storage requires scalability in file systems and also closeness to the machine, for which C++ is preferable.

C++'s scalability and rich set of libraries provide developers with the high concurrence and load tolerance which are the prerequisites for such scenarios.

8. Games

As C++ is closer to the hardware, its speed and faster execution help game developers utilize the CPU for thorough and intensive purposes. C++ removes the complexities of 3D in-game graphics. It also helps to optimize resource management which enables a multiplayer communication system.

9. Compilers

C++ being a low-level language is closer to the hardware. C++ also helped in developing a range of compilers including Dev-C++, Apple C++, Clang C++, and MINGW.